Understanding human behaviour at Police stations.

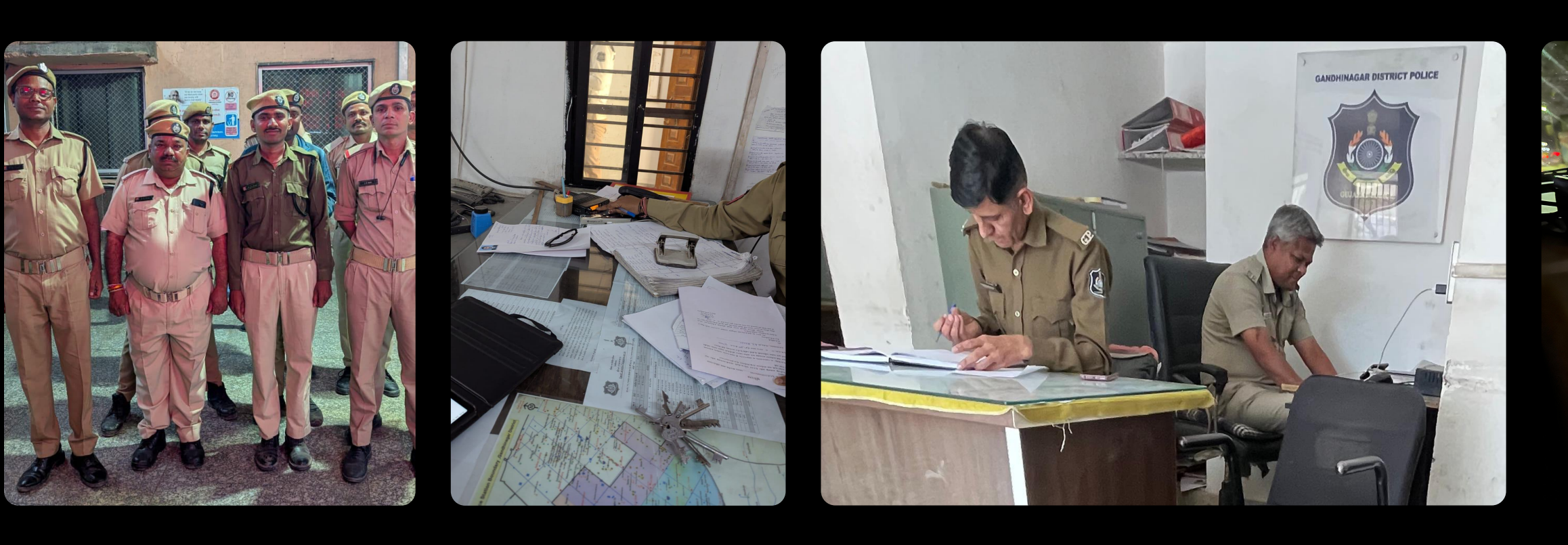
This case study examines how continuous wireless and mobile interruptions shape human behaviour, attention, and emotional response within police stations.

Project Overview

Challenge
Every police station hears the same wireless broadcast. At rush hour, it keeps interrupting the officer at the front desk while people wait to be heard

Approach
We watched, shadowed and tested. Out of 240+ behaviours, we chose the few that mattered most and found what causes them.

Outcome
A three-level plan: fix the source of the noise, the officer's workflow, and how the complainant is prepared.



Research Methodology

01.
Field Immersion & Context

Area of inquiry: How the front desk, the wireless, and station workflow actually run during rush hours

- Slow Ethnography
- Conversations
- Contextual inquiry
- Fly on the Wall
- Shadowing
- Field notes & conversations
- Articulation & categorisation

02.
Behavioural Mapping & Prioritisation

Area of inquiry: How the front desk, the wireless, and station workflow actually run during rush hours

- Stimulus mapping
- Spatial mapping
- Comparative field study
- Prioritisation

03.
Validation & Intervention Design

Area of inquiry: Whether the prioritised behaviours hold under simulated conditions, and which root causes to design for

- 7 simulations
- Appreciative inquiry
- Fishbone analysis
- Constraints mapping
- How Might We

The story Time

A visitor meets the radio before the officer

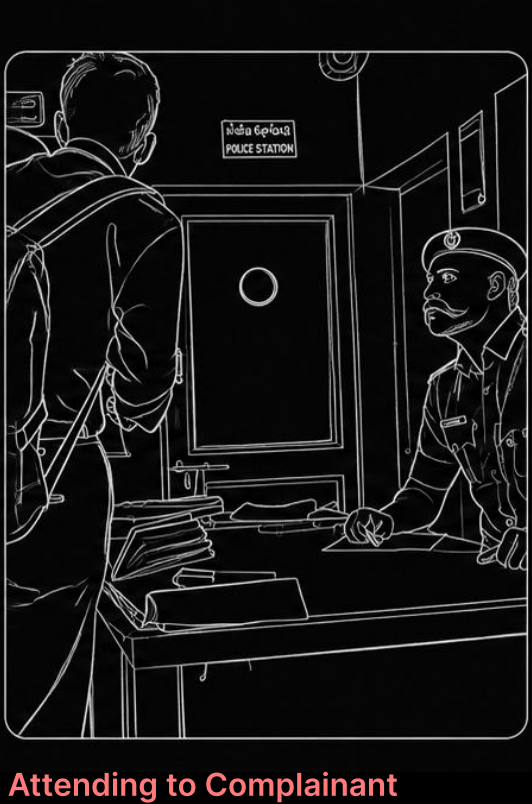
In a police station, the front desk officer(PSO) is the first person a visitor meets.

But the PSO is also the person who **listens to the wireless**. Messages come from the control room all day, to every station at once. So a person who comes with a problem often finds the officer listening to something else.

Recording Data and checking info during call and broadcast

Reducing the Volume of the Wireless while Attending Call

Responding to Wireless while on Call



Attending to Complainant during Broadcast

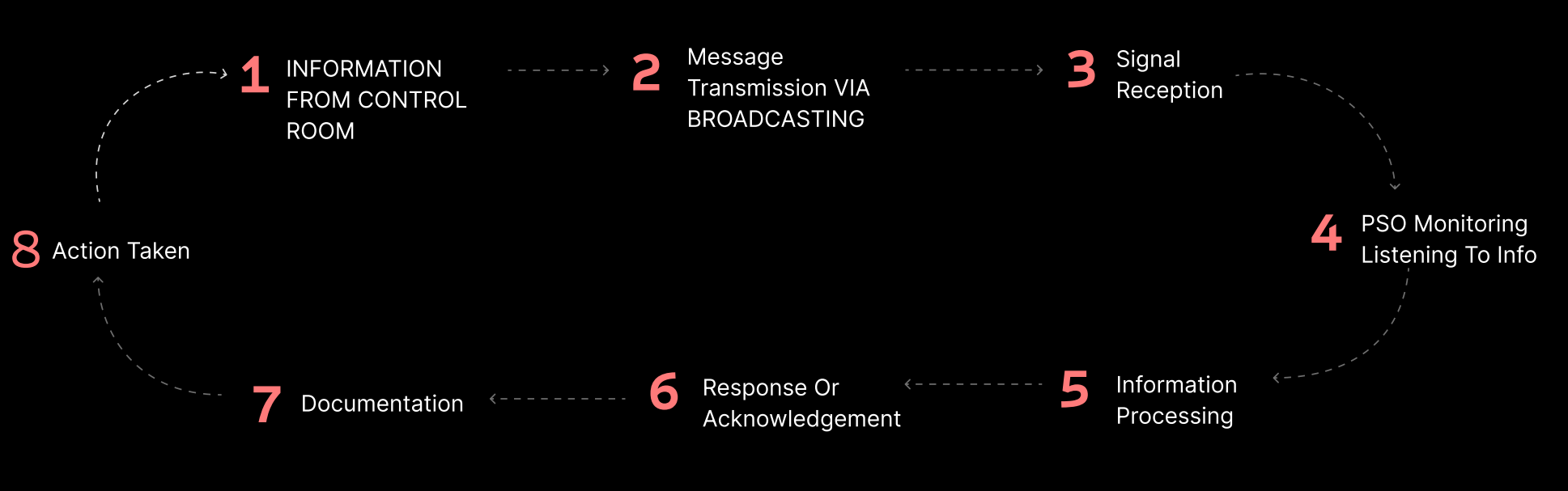
Zooming out: how a message travels

A message starts at the State control room. The District control room broadcasts it to all police stations. It reaches the PSO's wireless set, who must listen, understand, reply, write it down and act.

That is 8 steps, and the PSO does them while serving people.

STATE CONTROL ROOM → DISTRICT CONTROL ROOM → POLICE STATION WIRELESS AREA (PSO) → STATE & DISTRICT WORKSHOP

The wireless communication flow, from control room to action taken.

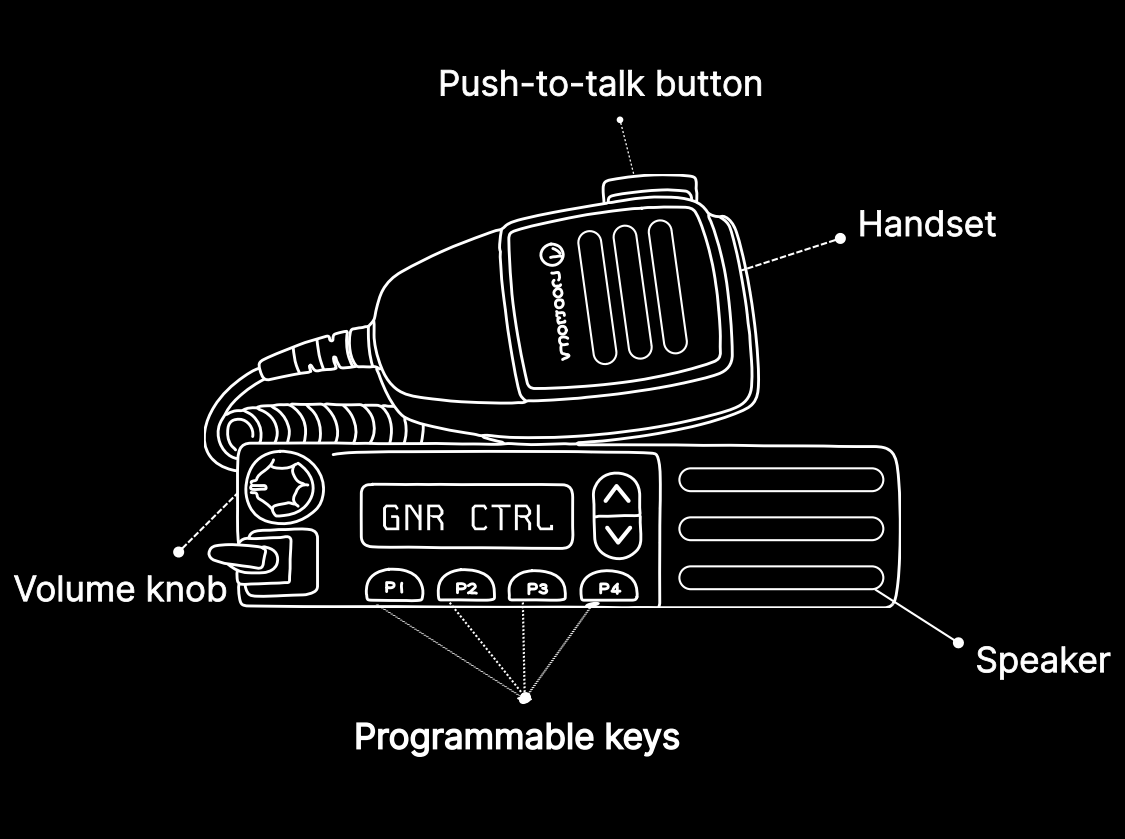

1 INFORMATION FROM CONTROL ROOM
2 Message Transmission VIA BROADCASTING
3 Signal Reception
4 PSO Monitoring Listening To Info
5 Information Processing
6 Response Or Acknowledgement
7 Documentation
8 Action Taken

The set itself shapes the behaviour

The set is **old and analog.** It has a push-to-talk button, a volume knob and a speaker. It plays every **broadcast aloud**, with **crackling noise** and no filter.

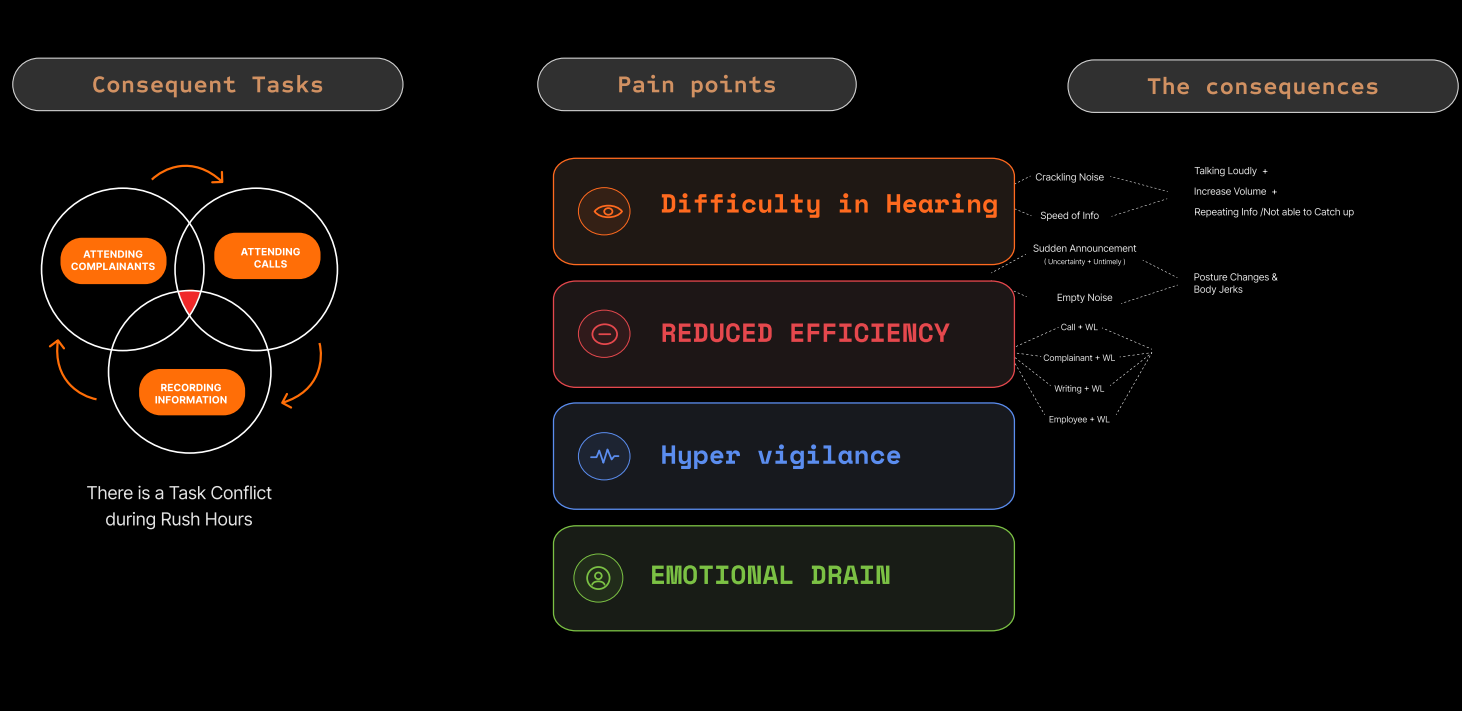
The officer cannot choose what to hear. Officers use it as they always have, so no one questions the design.

The wireless set: handset, volume knob, speaker and programmable



In rush hours, a broadcast lands on top of three tasks: attending complainants, taking calls and writing records. This is a task conflict. Hard to hear, always alert and switching tasks all the time, the officer talks loudly, repeats things, makes errors and gets tense.

Zooming in: what breaks at the desk



The cost is human

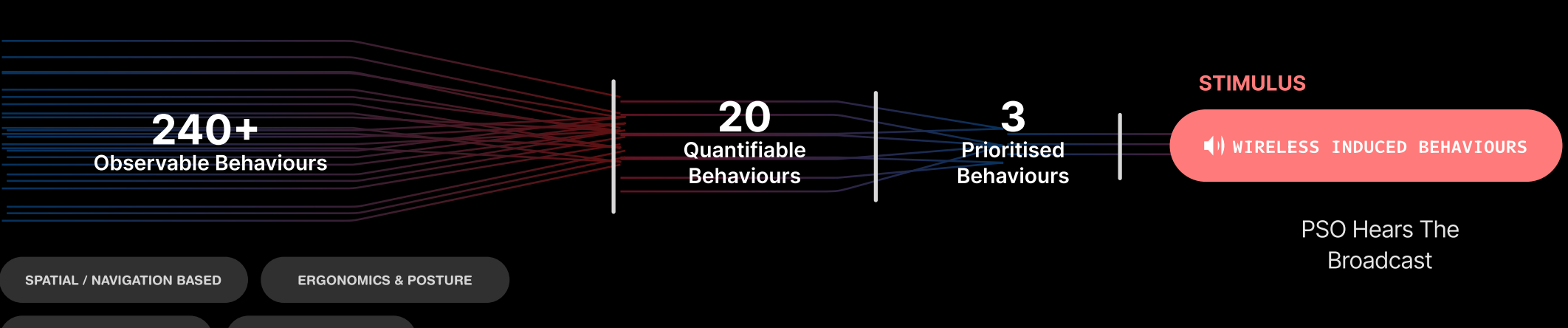
The complainant
stops speaking when the radio calls, and waits again when the officer pauses.

The officer
misses messages, writes slowly and makes errors under divided attention.

The desk
builds tension. People come already worried and leave unsure.

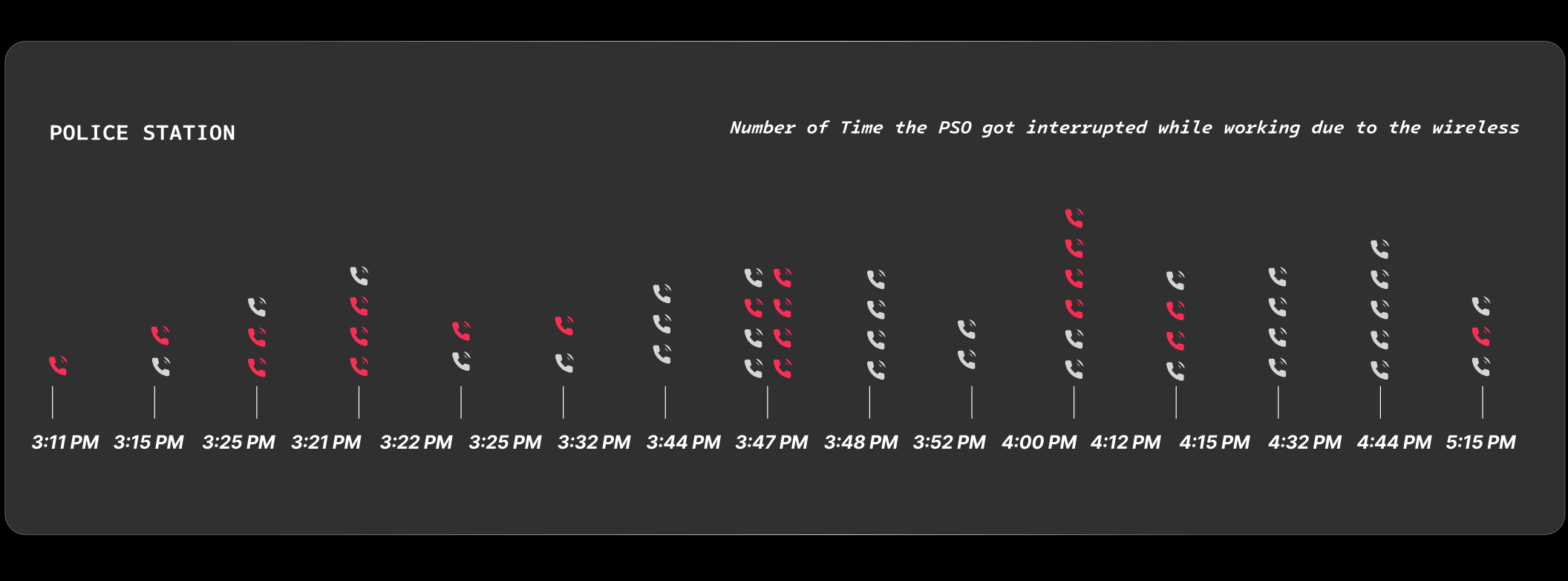
How We reached till here

We listed every behaviour we saw (240+), kept the 20 we could count, then chose 3 pain points to solve: difficulty hearing, hyper vigilance and task switching. We chose by asking which ones the wireless causes directly, and which hurt the officer and the complainant most.


240+ Observable Behaviours
20 Quantifiable Behaviours
3 Prioritised Behaviours
STIMULUS
WIRELESS INDUCED BEHAVIOURS
PSO Hears The Broadcast

- SPATIAL / NAVIGATION BASED
- ERGONOMICS & POSTURE
- WAITING BEHAVIOUR
- REPORTING COMPLAINTS
- PASSPORT VERIFICATION & RELATED TASKS
- INVESTIGATION TASKS
- TECHNICAL SUPPORT
- TECHNICAL SUPPORT

The data: interruptions come in clusters



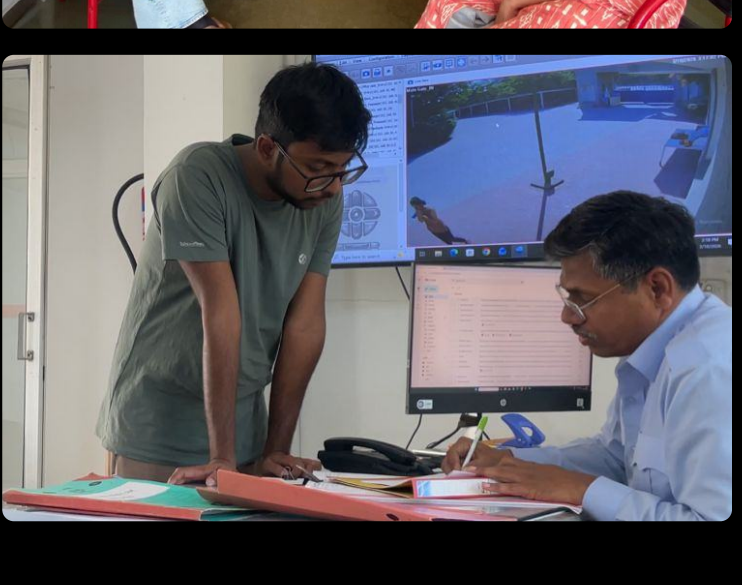
At one station we logged each time the wireless interrupted the PSO from 3:11 PM to 5:15 PM. Interruptions did not come evenly. They bunched together, and this is when mistakes and tension rise.

Testing it: 7 simulations at the main gate

We scored each simulation on six things: communication clarity, interruptions, task switching, information overload, urgency and responsibility, and efficiency.

7
Total Number of Simulations

85-90dB.
Noise at the station measured

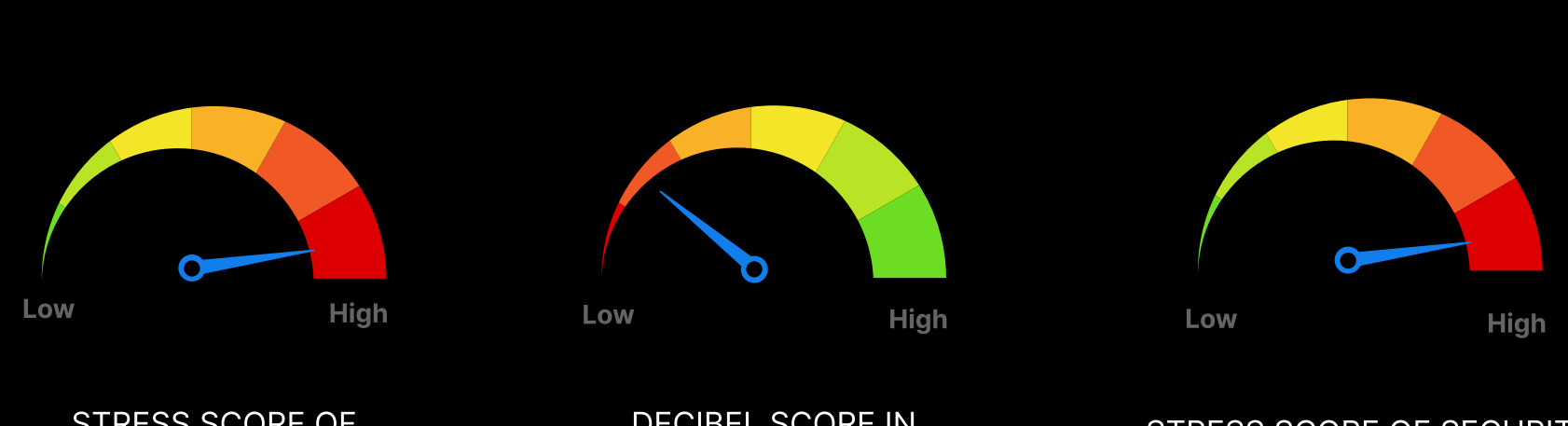


Observations :

- Intercom ringing repeatedly interrupted the PSO while interacting with the complainant, forcing them to stop the conversation and attend to the broadcast.
- Complainants stopped speaking multiple times whenever the intercom rang or when the PSO paused to attend to it.
- Participants reduced the speaker volume or physically moved the speaker aside to cope with the disturbance.
- PSO missed responding to messages from the intercom.
- The PSO had to listen to the intercom, talk to the complainant, answer phone calls, and write notes simultaneously, which made it difficult to focus on any single task.
- PSO struggled to continue writing or noting down the complainant's details, which slowed down the complaint-recording process.

Recorded Parameters

- COMMUNICATION CLARITY
- INTERRUPTIONS
- TASK SWITCHING
- INFORMATION OVERLOAD
- URGENCY & RESPONSIBILITY
- EFFICIENCY



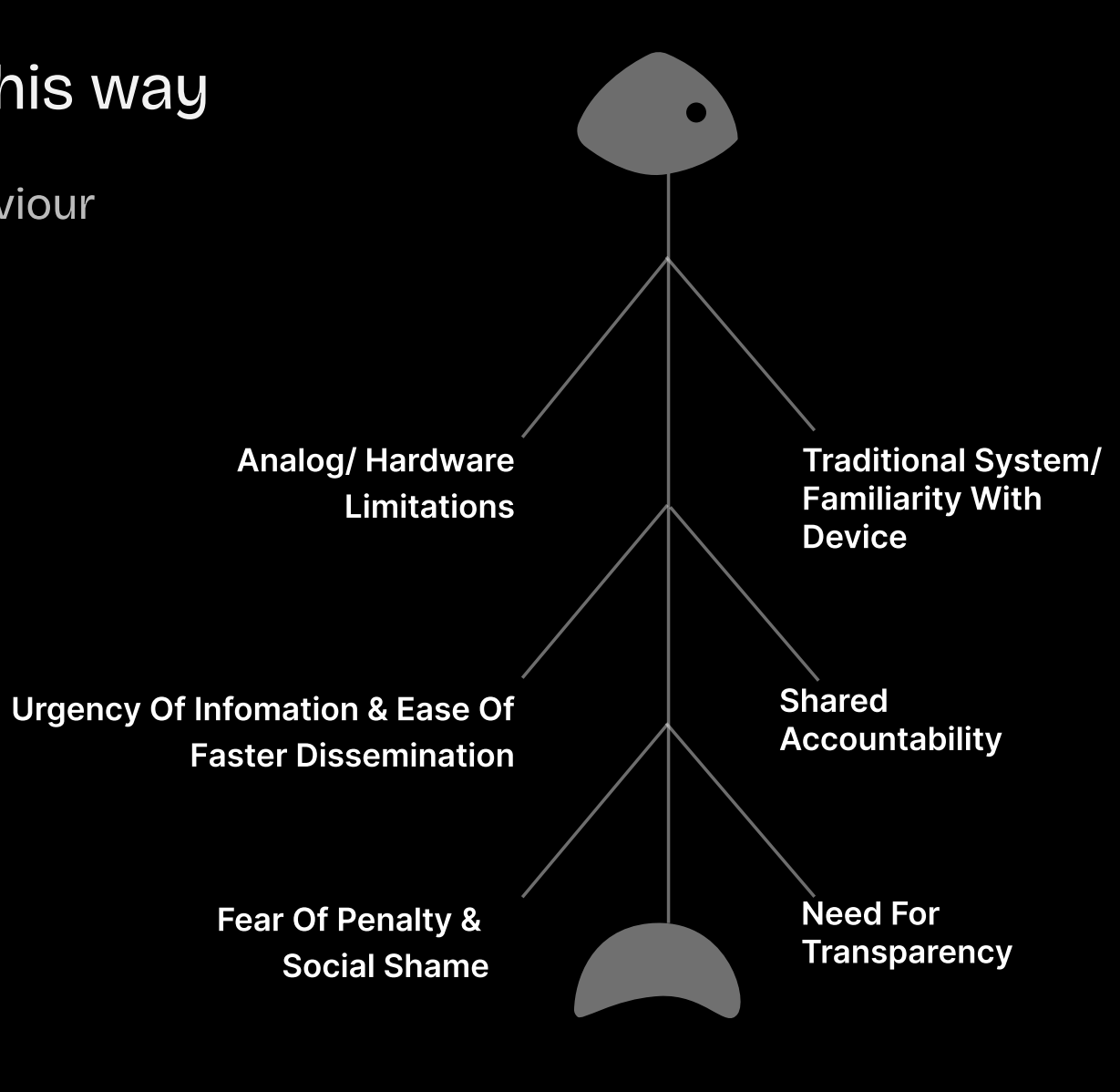
STRESS SCORE OF STATIONED PSO
(90)

DECIBEL SCORE IN POLICE STATION
(85-90)

STRESS SCORE OF SECURITY DURING SIMULATION
(90)

Why people behave this way

Six forces sit behind the behaviour


Analog/ Hardware Limitations
Traditional System/ Familiarity With Device
Urgency Of Infomation & Ease Of Faster Dissemination
Shared Accountability
Fear Of Penalty & Social Shame
Need For Transparency